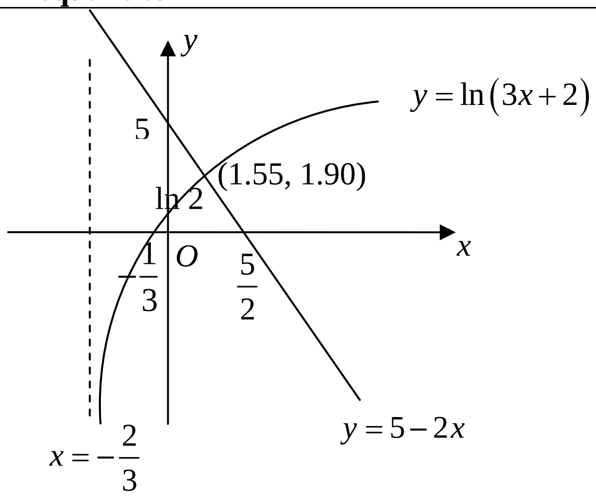
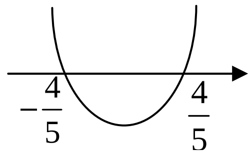
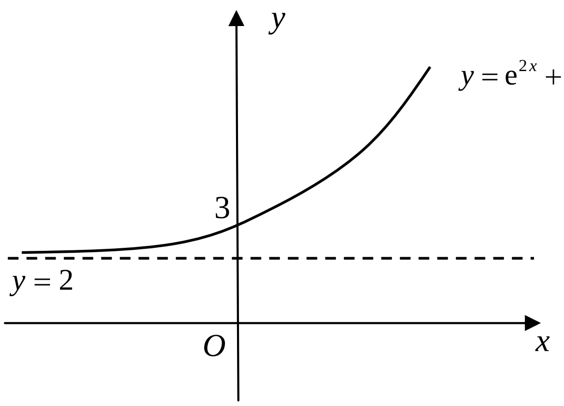
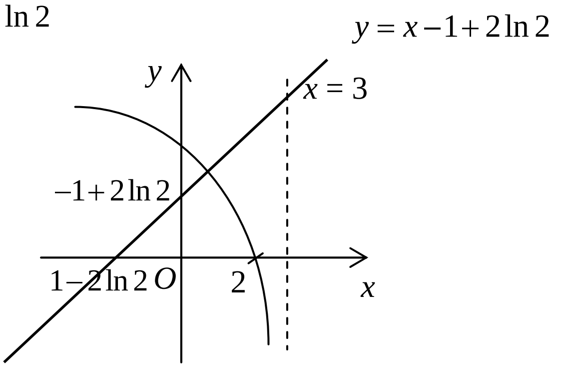
2013 H1 MATH (8864) JC 2 PRELIMINARY EXAM PAPER - SUGGESTED SOLUTIONS

Qn	Solution
1	**Logarithmic and Exponential functions**
	$3(\ln x)^2 = 4 + 4\ln\left(\frac{1}{x}\right)$ $\Rightarrow 3(\ln x)^2 = 4 - 4\ln x$ Let $u = \ln x$. $\Rightarrow 3u^2 = 4 - 4u$ $\Rightarrow 3u^2 + 4u - 4 = 0$ (Use of GC is allowed) $\Rightarrow (3u-2)(u+2) = 0$ $\Rightarrow u = \frac{2}{3}$ or -2 $\Rightarrow \ln x = \frac{2}{3}$ or $\ln x = -2$ $\Rightarrow x = e^{\frac{2}{3}}$ or $x = e^{-2}$ (Exact ans: Use of GC is not allowed)

Qn	Solution
2	**Inequalities**
(a)	 $5 - 2x > \ln(3x+2)$ Using GC, the 2 graphs intersect at x = 1.55. $\therefore -\frac{2}{3} < x < 1.55$
(b)	$2x^2 + 5px + 2 = 0$ For no reals roots, Discriminant < 0

	$(5p)^2-4(2)(2)<0$ $25p^2-16<0$ $(5p+4)(5p-4)<0$  $\therefore -\frac{4}{5}<p<\frac{4}{5}$

Qn	Solution
3	**Integration Techniques and Application of Integration**
(a)	$\int \frac{1}{2-3x}+\sqrt{3x+1}\ \mathrm{d}x$ $=\frac{1}{-3}\ln\lvert 2-3x\rvert+\frac{(3x+1)^{\frac{3}{2}}}{\frac{3}{2}(3)}+C$ $=-\frac{1}{3}\ln\lvert 2-3x\rvert+\frac{2(3x+1)^{\frac{3}{2}}}{9}+C$
(b)(i)	
(b)(ii)	Area $=\int_0^2 \mathrm{e}^{2x}+2\ \mathrm{d}x$ $=\left[\frac{\mathrm{e}^{2x}}{2}+2x\right]_0^2$ $=\frac{\mathrm{e}^4}{2}+4-\frac{1}{2}$ $=\frac{\mathrm{e}^4+7}{2}$ units^2

Qn	Solution
4	**Graphing Techniques (include tangent/normal)**
(i)	$y = 2\ln(3-x)$ $\frac{dy}{dx} = -\frac{2}{3-x}$
(ii)	When $x=1$, $y = 2\ln(3-1) = 2\ln 2$, $\frac{dy}{dx} = -1$ Equation of the normal at point *D*, $y - 2\ln 2 = 1(x-1)$ $y = x - 1 + 2\ln 2$
(iii)	At *F*, when $x=0$ $y = 0 - 1 + 2\ln 2 = 2\ln 2 - 1$ At E, when $y=0$ $0 = x - 1 + 2\ln 2$ $x = 1 - 2\ln 2$  Area of triangle $= \frac{1}{2}(2\ln 2 - 1)(2\ln 2 - 1)$ $= 0.0746 \text{ units}^2$

<table>
<tr><th>Qn</th><th>Solution</th></tr>
<tr><td>5</td><td>Application of Differentiation</td></tr>
<tr><td>(i)</td><td>

$y = 4\sqrt{10}x - x^2$

$\frac{\mathrm{d}y}{\mathrm{d}x} = 4\sqrt{10} - 2x$

When y is maximum, $\frac{\mathrm{d}y}{\mathrm{d}x} = 0.$

$\Rightarrow x = 2\sqrt{10}$

<table>
<tr><td>x</td><td>$2\sqrt{10}^-$</td><td>$2\sqrt{10}$</td><td>$2\sqrt{10}^+$</td></tr>
<tr><td>$\frac{\mathrm{d}y}{\mathrm{d}x}$</td><td>/</td><td>—</td><td>\</td></tr>
</table>

Maximum height reached $= 4\sqrt{10}\left(2\sqrt{10}\right) - \left(2\sqrt{10}\right)^2$

$= 40\text{m}$

</td></tr>
<tr><td>(ii)</td><td>

$\frac{\mathrm{d}y}{\mathrm{d}t} = 40 - 20t$

At $t = 0$,

Rate of change of height $= 40\text{ms}^{-1}$

</td></tr>
<tr><td>(iii)</td><td>

At $t = 0$,

$y = 0,\ x = 0$ and $\frac{\mathrm{d}y}{\mathrm{d}x} = 4\sqrt{10}.$

At $t = 0$,

Rate of change of horizontal distance travelled, $\frac{\mathrm{d}x}{\mathrm{d}t} = \frac{\mathrm{d}x}{\mathrm{d}y} \times \frac{\mathrm{d}y}{\mathrm{d}t}$

$= \frac{1}{4\sqrt{10}} \times (40)$

$\approx 3.16\text{ms}^{-1} \quad (3 \text{ s.f.})$

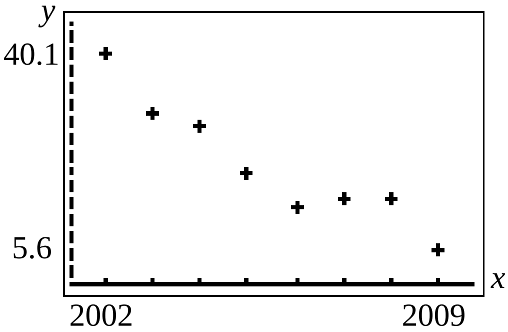
</td></tr>
</table>

Section B

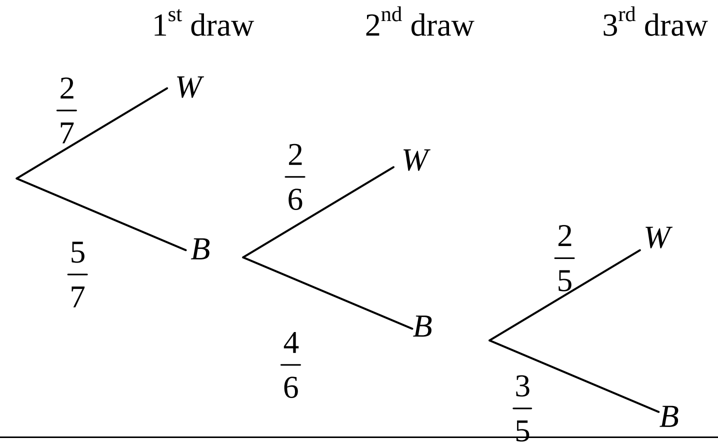
Qn	Solution
6	**Sampling Methods**
(i)	It is too costly or time consuming to test all the mugs that he produced.
(ii)	No, it is because mugs from different sizes have different probabilities of being selected since the proportion of each mug size is different.
(iii)	Stratified sampling. Select the following number of mugs **randomly** for each size: Size 1: $\frac{2}{5}\times 40=16$ mugs Size 2: $\frac{3}{5}\times 40=24$ mugs

Qn	Solution
7	**Hypothesis Testing**
(i)	Let X be the mass of a bag of Easy bake bread mix (in grams). Let μ be the population mean mass of a randomly chosen bag of Easy-bake bread mix (in grams). $H_0: \mu = 400$ $H_1: \mu \neq 400$ Given: $X \sim \mathrm{N}\left(\mu,\sigma^2\right)$, $\therefore \overline{X} \sim \mathrm{N}\left(\mu,\frac{\sigma^2}{n}\right)$ Under H_0, $\overline{X} \sim \mathrm{N}\left(400,\frac{5^2}{50}\right)$. Test statistic: $Z=\frac{\overline{X}-\mu}{\sigma/\sqrt{n}}$ Level of significance: 5% Reject H_0 if z-value < –1.9600 or z-value > 1.9600. Since H_0 is rejected at 5% level, z-value = $\frac{\overline{x}-400}{5/\sqrt{50}} < -1.9600$ or $\frac{\overline{x}-400}{5/\sqrt{50}} > 1.9600$ $\Rightarrow \overline{x} < 398.614$ or $\overline{x} > 401.386$ Set of values of $\overline{x} = \left\{\overline{x} \in \mathbb{R} : \overline{x} < 398.6 \text{ or } \overline{x} > 401.4\right\}$
(ii)	Since $\overline{x} = 399$, thus $398.6 < \overline{x} < 401.4$. Hence we do not reject H_0 and conclude that there is insufficient evidence, at 5% level of significance, that the mean mass is not 400 grams.
(iii)	"at the 5% level of significance" means there is a probability of 0.05 of concluding that the mean mass of a bag of Easy-bake bread mix is not 400 grams when in fact it is 400 grams.

(iv)	It is not necessary to assume normal distribution since the sample size n = 50 is large so by Central Limit Theorem, the sample mean $\overline{X}$ follows a normal distribution approximately. Alternative: It is not necessary to assume normal distribution since the sample size n = 50 is large so by Central Limit Theorem, $\overline{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ approximately.

Qn8	Solution
8	**Correlation & Regression**
(i)	
(ii)	product moment correlation coefficient, $r = -0.946$ (3 s.f.)
(iii)	The regression line of y on x is $y = 8622.55 - 4.2893x$ $y = 8622.55 - 4.29x$ The regression line of x on y is $x = 2009.8 - 0.20863y$ $x = 2010 - 0.209y$
(iv)	Hence, when $y = 1$, $1 = 8622.55 - 4.2893x$ $x = 2010.01$ In year 2010, the average cost of COE drops to one thousand dollars. This estimate is not valid because y = 1 is not within the data range of y.
(v)	The value of the gradient means that the average cost of COE drops by about $4,290 every year.
(vi)	A linear model would not be appropriate because in the long run, the cost of COE will be negative.

Qn	Solution
9	**Normal and Sampling**
(a)	Let X be the mass of a pear in grams. $X \sim N(\mu, \sigma^2)$ $\mathrm{P}(X > 200) = 0.15$ $\mathrm{P}\left(Z > \frac{200-\mu}{\sigma}\right) = 0.15$ $\mathrm{P}\left(Z \le \frac{200-\mu}{\sigma}\right) = 0.85$ $\frac{200-\mu}{\sigma} = 1.0364$ $200-\mu = 1.0364\sigma \quad ----- (1)$ $\mathrm{P}(X < 135) = 0.25$ $\mathrm{P}\left(Z < \frac{135-\mu}{\sigma}\right) = 0.25$ $\frac{135-\mu}{\sigma} = -0.67449$ $135-\mu = -0.67449\sigma \quad ----- (2)$ Using GC, $\sigma = 37.992 = 38.0$ (3 s.f.) $\mu = 160.63$ $= 161$ (3 s.f)
(b) (i)	Let A be the mass an apple in kg. Let B be the mass a plum in kg. $A \sim \mathrm{N}(0.155, 0.03^2)$ $\quad B \sim \mathrm{N}(0.025, 0.005^2)$ Let $Y = A_1 + A_2 + (B_1 + B_2 + B_3)$ $Y \sim \mathrm{N}(0.385, 0.001875)$ $\mathrm{P}(Y < 0.36) = 0.28185$ $= 0.282$ (3s.f)
(ii)	Let $W = 4(A_1 + A_2 + A_3) + 12(B_1 + B_2 + B_3)$. Let $T = 12(B_1 + B_2 + \ldots + B_{10})$. $W \sim \mathrm{N}(2.76, 0.054)$ and $T \sim \mathrm{N}(3, 0.036)$ $T - W \sim \mathrm{N}(0.24, 0.09)$ $\mathrm{P}(T - W \ge 0.5) = 0.193$ (3s.f)

Qn	Solution
10	**Probability**
(a)	Given that $\mathrm{P}(A)=\dfrac{9}{20}$, $\mathrm{P}(A\mid B)=\dfrac{6}{13}$ and $\mathrm{P}(A\cap B)=\dfrac{3}{10}$. $\mathrm{P}(A\mid B)=\dfrac{\mathrm{P}(A\cap B)}{\mathrm{P}(B)}$ $\dfrac{6}{13}=\dfrac{\frac{3}{10}}{\mathrm{P}(B)}$ $\mathrm{P}(B)=\dfrac{\frac{3}{10}}{\frac{6}{13}}$ $\mathrm{P}(B)=\dfrac{13}{20}$. $\mathrm{P}(A'\cap B')=1-\mathrm{P}(A\cup B)$ $\mathrm{P}(A'\cap B')=1-\left[\mathrm{P}(A)+\mathrm{P}(B)-\mathrm{P}(A\cap B)\right]$ $\mathrm{P}(A'\cap B')=1-\left(\dfrac{9}{20}+\dfrac{13}{20}-\dfrac{3}{10}\right)$ $=1-\dfrac{4}{5}$ $=\dfrac{1}{5}$ $\mathrm{P}(A\mid B)=\dfrac{6}{13}$ $\mathrm{P}(A)=\dfrac{9}{20}$ Since $\mathrm{P}(A\mid B)\neq \mathrm{P}(A)$, A and B are not independent.
(bi)	

(ii)	Required probability $= \mathrm{P}(W) + \mathrm{P}(BW) + \mathrm{P}(BBW)$ $= \frac{2}{7} + \left(\frac{5}{7}\right)\left(\frac{2}{6}\right) + \left(\frac{5}{7}\right)\left(\frac{4}{6}\right)\left(\frac{2}{5}\right)$ $= \frac{5}{7}$
(iii)	$\mathrm{P}(\text{Elizabeth draws exactly 3 balls} \mid \text{draws fewer than 4 throws})$ $= \frac{\mathrm{P}(\text{Elizabeth draws exactly 3 balls})}{\mathrm{P}(\text{fewer than 4 throws})}$ $= \frac{\left(\frac{5}{7}\right)\left(\frac{4}{6}\right)\left(\frac{2}{5}\right)}{\frac{5}{7}}$ $= \frac{4}{15}$

Qn	Solution
11	**Binomial distribution (incl. conditions of Bino, approx. normal)**
(i)	- Whether a person has gene A is independent of any other person having the gene A. - The probability that a person has the gene A must be constant throughout the sample.
(ii)	$X \sim \mathrm{B}(n, 0.37)$ When $n = 20$, $\quad \mathrm{P}(X \le 10) = 0.92246 > 0.9$ $n = 21$, $\quad \mathrm{P}(X \le 10) = 0.89008 < 0.9$ $\therefore$ greatest $n = 20$
(iii)	$X \sim \mathrm{B}(10, 0.37)$ $\mathrm{P}(3 < X \le 7) = \mathrm{P}(X \le 7) - \mathrm{P}(X \le 3)$ $= 0.533$
(iv)	$X \sim \mathrm{B}(200, 0.37)$ $\mathrm{P}(X = 73) = 0.0579$ $\mathrm{P}(X = 74) = 0.0584$ $\mathrm{P}(X = 75) = 0.0576$ Hence, the most probable number is 74 people.
(v)	$X \sim \mathrm{B}(200, 0.37)$ Since np = 200(0.37) = 74 > 5 and nq = 200(0.63) = 126 > 5, then $X \sim \mathrm{N}(74, 46.62)$ approximately. $\mathrm{P}(X > 80) = \mathrm{P}(X > 80.5)$ after continuity correction $= 0.17055$ $= 0.171$ (3s.f.)
(vi)	Let Y be the number of samples, out of 5, that have more than 80 people with gene A. $Y \sim \mathrm{B}(5, 0.17055)$ $\mathrm{P}(Y = 3) = 0.0341$ (3s.f.)